Customize your configuration

SnapCenter Plug-in for VMware vSphere

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To improve operational efficiency, you can modify the `scbr.override` configuration file to change default values. These values control settings such as the number of VMware snapshots that are created or deleted during a backup or the amount of time before a backup script stops running.

The `scbr.override` configuration file is used by the SnapCenter Plug-in for VMware vSphere environments that support SnapCenter application-based data protection operations. If this file does not exist, then you must create it from the template file.

Create the scbr.override configuration file

1. Go to `/opt/netapp/scvservice/standalone_aegis/etc/scbr/scbr.override-template`.
2. Copy the `scbr.override-template` file to a new file called `scbr.override` in the `\opt\netapp\scvservice\standalone_aegis\etc\scbr` directory.

Properties you can override

- By default, the template uses hash symbol to comment the configuration properties. To use a property to modify a configuration value, you must remove the `#` characters.
- You must restart the service on the SnapCenter Plug-in for VMware vSphere host for the changes to take effect.

You can use the following properties that are listed in the `scbr.override` configuration file to change default values.

- **dashboard.protected.vm.count.interval=7**

  Specifies the number of days for which the dashboard displays VM protection status.

  The default value is "7".

- **guestFileRestore.guest.operation.interval=5**

  Specifies the time interval, in seconds, that SnapCenter Plug-in for VMware vSphere monitors for completion of guest operations on the guest (Online Disk and Restore Files). The total wait time is set by `guestFileRestore.online.disk.timeout` and `uestFileRestore.restore.files.timeout`.

  The default value is "5".

- **guestFileRestore.monitorInterval=30**

  Specifies the time interval, in minutes, that the SnapCenter VMware plug-in monitors for expired guest file restore sessions. Any session that is running beyond the configured session time is disconnected.

  The default value is "30".

- **guestFileRestore.online.disk.timeout=100**

  Specifies the time, in seconds, that the SnapCenter VMware plug-in waits for an online disk operation on a

guest VM to complete. Note that there is an additional 30-second wait time before the plug-in polls for completion of the online disk operation.

The default value is "100".

- **guestFileRestore.restore.files.timeout=3600**

  Specifies the time, in seconds, that the SnapCenter VMware plug-in waits for a restore files operation on a guest VM to complete. If the time is exceeded, the process is ended and the job is marked as failed.

  The default value is "3600" (1 hour).

- **guestFileRestore.robocopy.directory.flags=/R:0 /W:0 /ZB /CopyAll /EFSRAW /A-:SH /e /NJH /NDL /NP**

  Specifies the extra robocopy flags to use when copying directories during guest file restore operations.

  Do not remove `/NJH` or add `/NJS` because this will break the parsing of the restore output.

  Do not allow unlimited retries (by removing the `/R` flag) because this might cause endless retries for failed copies.

  The default values are `"/R:0 /W:0 /ZB /CopyAll /EFSRAW /A-:SH /e /NJH /NDL /NP"` .

- **guestFileRestore.robocopy.file.flags=/R:0 /W:0 /ZB /CopyAll /EFSRAW /A-:SH /NJH /NDL /NP**

  Specifies the extra robocopy flags to use when copying individual files during guest file restore operations.

  Do not remove `/NJH` or add `/NJS` because this will break the parsing of the restore output.

  Do not allow unlimited retries (by removing the `/R` flag) because this might cause endless retries for failed copies.

  The default values are `"/R:0 /W:0 /ZB /CopyAll /EFSRAW /A-:SH /NJH /NDL /NP"`.

- **guestFileRestore.sessionTime=1440**

  Specifies the time, in minutes, that SnapCenter Plug-in for VMware vSphere keeps a guest file restore session active.

  The default value is "1440" (24 hours).

- **guestFileRestore.use.custom.online.disk.script=true**

  Specifies whether to use a custom script for onlining disks and retrieving drive letters when creating guest file restore sessions. The script must be located at `[Install Path]\etc\guestFileRestore_onlineDisk.ps1`. A default script is provided with the installation. The values `[Disk_Serial_Number]`, `[Online_Disk_Output]`, and `[Drive_Output]` are replaced in the script during the attach process.

  The default value is "false".

- **include.esx.initiator.id.from.cluster=true**

  Specifies that the SnapCenter VMware plug-in should include iSCSI and FCP initiator IDs from all the ESXi hosts in the cluster in the application over VMDK workflows.

The default value is "false".

- **max.concurrent.ds.storage.query.count=15**

  Specifies the maximum number of concurrent calls that the SnapCenter VMware plug-in can make to the SnapCenter Server to discover the storage footprint for the datastores. The plug-in makes these calls when you restart the Linux service on the SnapCenter VMware plug-in VM host.

- **nfs.datastore.mount.retry.count=3**

  Specifies the maximum number of times the SnapCenter VMware plug-in tries to mount a volume as a NFS Datastore in vCenter.

  The default value is "3".

- **nfs.datastore.mount.retry.delay=60000**

  Specifies the time, in milliseconds, that the SnapCenter VMware plug-in waits between attempts to mount a volume as a NFS Datastore in vCenter.

  The default value is "60000" (60 seconds).

- **script.virtual.machine.count.variable.name= VIRTUAL_MACHINES**

  Specifies the environmental variable name that contains the virtual machine count. You must define the variable before you execute any user-defined scripts during a backup job.

  For example, VIRTUAL_MACHINES=2 means that two virtual machines are being backed up.

- **script.virtual.machine.info.variable.name=VIRTUAL_MACHINE.%s**

  Provides the name of the environmental variable that contains information about the nth virtual machine in the backup. You must set this variable before executing any user defined scripts during a backup.

  For example, the environmental variable VIRTUAL_MACHINE.2 provides information about the second virtual machine in the backup.

- **script.virtual.machine.info.format= %s|%s|%s|%s|%s**

  Provides information about the virtual machine. The format for this information, which is set in the environment variable, is the following: `VM name|VM UUID| VM power state (on|off)|VM snapshot taken (true|false)|IP address(es)`

  The following is an example of the information you might provide:

  ```
  VIRTUAL_MACHINE.2=VM 1|564d6769-f07d-6e3b-68b1f3c29ba03a9a|POWERED_ON||true|10.0.4.2
  ```

- **storage.connection.timeout=600000**

  Specifies the amount of time, in milliseconds, that the SnapCenter Server waits for a response from the storage system.

  The default value is "600000" (10 minutes).

- **vmware.esx.ip.kernel.ip.map**

  There is no default value. You use this value to map the ESXi IP address to the VMkernel IP address. By default, the SnapCenter VMware plug-in uses the management VMkernel adapter IP address of the ESXi host. If you want the SnapCenter VMware plug-in to use a different VMkernel adapter IP address, you must provide an override value.

  In the following example, the management VMkernel adapter IP address is 10.225.10.56; however, the SnapCenter VMware plug-in uses the specified address of 10.225.11.57 and 10.225.11.58. And if the management VMkernel adapter IP address is 10.225.10.60, the plug-in uses the address 10.225.11.61.

  ```
  vmware.esx.ip.kernel.ip.map=10.225.10.56:10.225.11.57,10.225.11.58;
  10.225.10.60:10.225.11.61
  ```

- **vmware.max.concurrent.snapshots=30**

  Specifies the maximum number of concurrent VMware snapshots that the SnapCenter VMware plug-in performs on the server.

  This number is checked on a per datastore basis and is checked only if the policy has "VM consistent" selected. If you are performing crash-consistent backups, this setting does not apply.

  The default value is "30".

- **vmware.max.concurrent.snapshots.delete=30**

  Specifies the maximum number of concurrent VMware snapshot delete operations, per datastore, that the SnapCenter VMware plug-in performs on the server.

  This number is checked on a per datastore basis.

  The default value is "30".

- **vmware.query.unresolved.retry.count=10**

  Specifies the maximum number of times the SnapCenter VMware plug-in retries sending a query about unresolved volumes because of "…time limit for holding off I/O…" errors.

  The default value is "10".

- **vmware.quiesce.retry.count=0**

  Specifies the maximum number of times the SnapCenter VMware plug-in retries sending a query about VMware snapshots because of "…time limit for holding off I/O…" errors during a backup.

  The default value is "0".

- **vmware.quiesce.retry.interval=5**

  Specifies the amount of time, in seconds, that the SnapCenter VMware plug-in waits between sending the queries regarding VMware snapshot "…time limit for holding off I/O…" errors during a backup.

  The default value is "5".

- **vmware.query.unresolved.retry.delay= 60000**

Specifies the amount of time, in milliseconds, that the SnapCenter VMware plug-in waits between sending the queries regarding unresolved volumes because of "…time limit for holding off I/O…" errors. This error occurs when cloning a VMFS datastore.

The default value is "60000" (60 seconds).

- **vmware.reconfig.vm.retry.count=10**

  Specifies the maximum number of times the SnapCenter VMware plug-in retries sending a query about reconfiguring a VM because of "…time limit for holding off I/O…" errors.

  The default value is "10".

- **vmware.reconfig.vm.retry.delay=30000**

  Specifies the maximum time, in milliseconds, that the SnapCenter VMware plug-in waits between sending queries regarding reconfiguring a VM because of "…time limit for holding off I/O…" errors.

  The default value is "30000" (30 seconds).

- **vmware.rescan.hba.retry.count=3**

  Specifies the amount of time, in milliseconds, that the SnapCenter VMware plug-in waits between sending the queries regarding rescanning the host bus adapter because of "…time limit for holding off I/O…" errors.

  The default value is "3".

- **vmware.rescan.hba.retry.delay=30000**

  Specifies the maximum number of times the SnapCenter VMware plug-in retries requests to rescan the host bus adapter.

  The default value is "30000".

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